Statistics

Outlier: A value whose attributes are significantly different from rest of the group

outlier

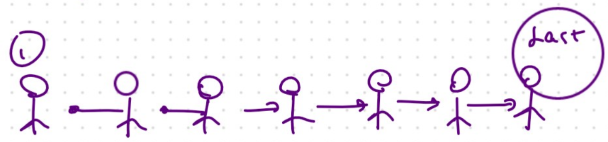
Ages = {10, 20, 40, 500}

Shapes = {circles, squares, triangle, star}

images = {dog1.jpg, dog2.jpg, cat1.jpg, cat2.jpg, lion1.jpg}

Measurement of dispersion

def: After finding the center, we now ask: how "SPREAD OUT" is the data from the center?



left — 2 hand
right — 4 hand

Why?

Player	Scores	Mean
A	50, 51, 49, 50, 52 ?	50.4
B	0, 110, 10, 90, 50 ?	52

reliable

Variance: the average of the squared distance of each data points from the mean

$$\{-1, 1, 2, -2\} = \frac{-1+1+2-2}{4} \quad \frac{0}{4} = 0$$

always guarantee's true value

$$(-1)^2 \ (2)^2 \Rightarrow 1, 4$$

$$\boxed{\sigma^2 = \frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}} \qquad \boxed{s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}$$

Ages = $\{2, 2, 4, 4\}$

$$\mu = \frac{2+2+4+4}{4} = 3$$

$i=0$

	x_i	μ	$(x_i-\mu)^2$
$i=1$	2	3	$(2-3)^2 = 1$
$i=2$	2	3	$(2-3)^2 = 1$
$i=3$	4	3	$(4-3)^2 = 1$
$i=4$	4	3	$(4-3)^2 = 1$

$$\frac{\sum_{i=1}^{N}(x_i-\mu)^2}{N} = \frac{4}{4} = 1$$

Variance = 1

$\boxed{\sigma^2 = 1}$

$\{10, 20, 30\}$

$$\bar{x} = \frac{60}{3} = 20$$

x_i	$(x_i-\bar{x})^2$	
10	$(10-20)^2$	$= 100$
20	$(20-20)^2$	$= 0$
30	$(30-20)^2$	$= 100$

$$s^2 = \frac{100+0+100}{3} = \frac{200}{3} \approx 66.67$$

The unit problem with variance:

Unit $\longrightarrow$ square unit

$x = \{\text{inches}\} \longrightarrow \text{inches}^2$

Standard Deviation: the square root of variance, to express in the same units as the original data

$\sigma = \sqrt{\text{population variance}}$

$s = \sqrt{\text{sample variance}}$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i-\mu)^2}{N}}$$

$s = \sqrt{66.7} \approx 8.16$

Verification:

$\{10, 20, 30\}$ $\mu = 20$

$\sigma = 8.16$

$20 - 8.16 = 11.84 \approx 10$

$20 + 8.16 = 28.16 \approx 30$

Break Down

Standard — base — average — mean

Deviation — how much each point differs from the base

(Pb1) A class of 5 students scored $\{10, 20, 30, 20, 40\}$

find mean, median & mode

(Pb2) Two cricketer's scores over 5 matches

Player A = $\{50, 51, 49, 50, 52\}$?

Player B = $\{0, 110, 10, 90, 150\}$?

μ

SD - ?

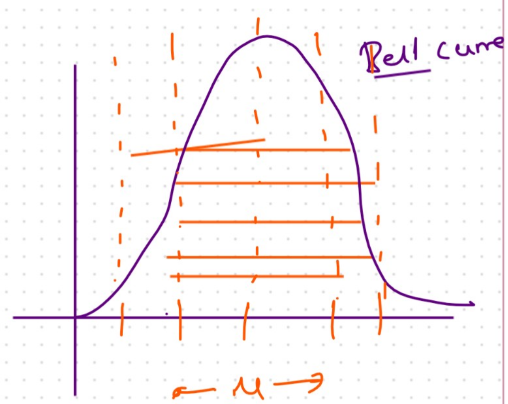
Choose one player, who is more consistent

outliers

$S.D << \mu$ $S.D >> \mu$ (X)

✓ $S.D \approx \mu$ (✓)

? does Variance & standard deviation is sensitive to outlier?



Misconceptions

(1) Mean & median are always similar

$\mu = 60$ median = 58

(2) Variance & SD measures different things

$D = \{1\ 2\ 4\ 2\} \rightarrow \{1\ 4\ 16\ 4\} \rightarrow \{1, 2, 4, 2\}$

lit → milli litre

kg → gram

③ Mode only works on categorical data

④ Higher mean = better performance ↓ 97

{40, 10, 20, 40, 60, 80}

✓ class A = 60 ↗ class A always has better students

class B = 40

↘ 40

without SD, mean alone is misleading ⟶ reason (outliers)

⑤ Categorical data can be averaged

{male, female} = $\frac{1+2}{2}$ = 1.5

↓ 1, ↓ 2

⑥ Ordinal data has equal intervals

Grade A ⟶ B

Grade B ⟶ C

not the size of gap b/w categories

Ⓠ Analyze house prices 30L, 35L, 2L, 32L, 500L, which measure should be reported by datascientist

mean = 125L

✓ median = 32L ⟶ accurately represent typical buyer's market

Price | Hallways | Squares | Area

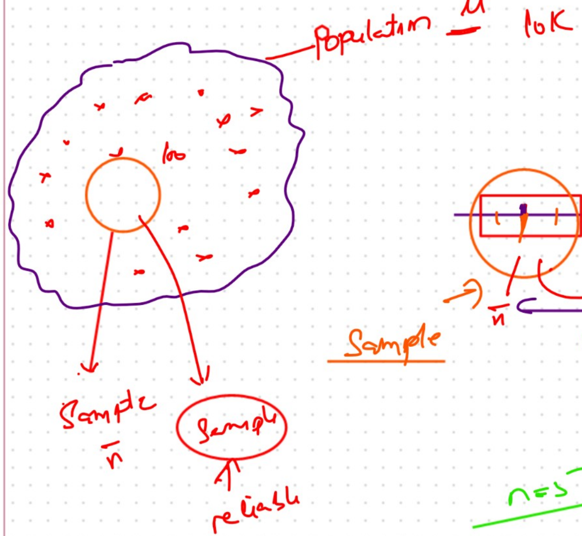
x_1 | x_1 | x_2 | x_3 | y

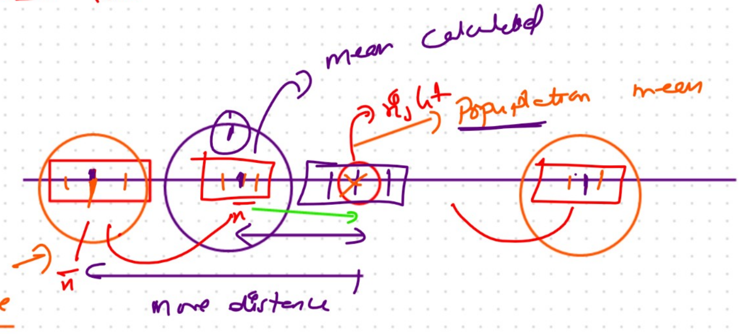
⟶ unknown

( ) ↗ reality

of current models

Q You are a data scientist at a company, you find that monthly revenue figures are 1.2Cr, 1.5Cr, 1.1Cr, 1.4Cr, 0.1Cr. The manager asks "Are our revenue consistent". How would you answer using statistics





$\bar{x} \approx \mu$

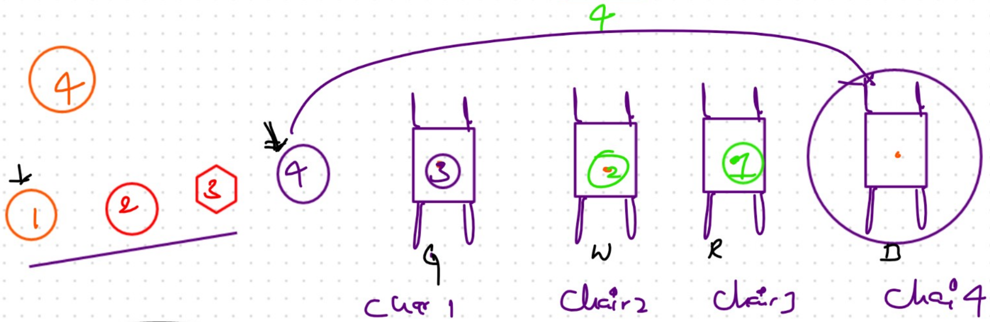
Ages

$\frac{1000}{50} = 20$

$\frac{1000}{4} = 250$

$n=5 \quad \frac{1000}{5} = 200$

$n=4 \quad \frac{1000}{4} = 250$



degree of freedom

Concept → Bascal correction

$(n-1)$